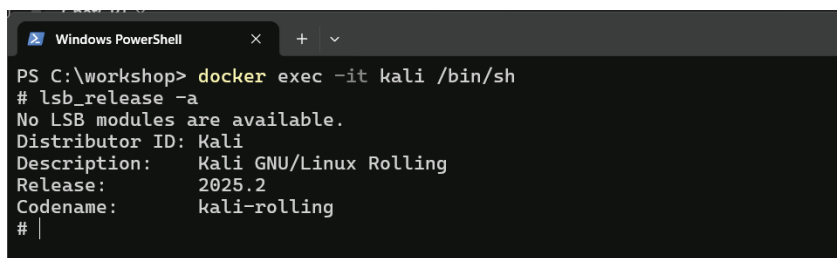


on the host is mapped to port 80 in the DVWA container, enabling you to access the web interface from your host machine. Isolated networking ensures that containers do not interfere with your host machine's network, providing a secure environment to practice penetration testing and security research.

Docker Compose simplifies the process of setting up a multi-container environment by allowing you to configure a custom network for all services to interact seamlessly. In this security test bed, all containers are connected to the pentest-network, ensuring that services like Kali Linux can reach Metasploitable2 for scanning, exploitation, and penetration testing exercises.

Networking is a critical aspect of any security test bed because it allows you to simulate real-world attack scenarios. For example: Kali Linux can scan for open ports and vulnerabilities on the Metasploitable2 container using tools like nmap. The DVWA web application can be targeted with attacks such as SQL injection or cross-site scripting (XSS), and security researchers can test their ability to detect and mitigate these attacks. Metasploitable2 provides vulnerable services that can be exploited using the Metasploit Framework running inside the Kali Linux container. This setup mimics the network interactions found in real-world penetration testing, where different systems and services are networked together, and security professionals need to identify and exploit vulnerabilities.

The Docker Compose file you've configured creates a flexible, multi-container security test bed, allowing you to experiment with various tools and techniques. By using Docker networking, the containers interact seamlessly within the isolated pentest-network, providing a realistic environment for penetration testing, web application security testing, and



```
Windows PowerShell
PS C:\workshop> docker exec -it kali /bin/sh
# lsb_release -a
No LSB modules are available.
Distributor ID: Kali
Description:    Kali GNU/Linux Rolling
Release:        2025.2
Codename:       kali-rolling
#
```

Figure 2: Checking the Kali container version

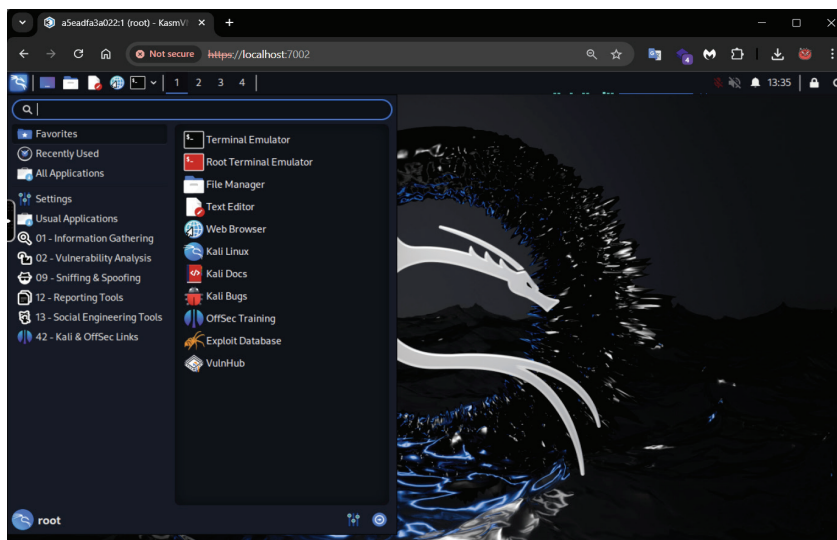


Figure 3: Kali GUI in Windows browser

exploit development. The simplicity of Docker Compose, combined with its powerful networking features, makes it an ideal choice for setting up a scalable, repeatable security lab environment.

Accessing the security environment

Now that your security test bed is up and running, you can interact with the containers either through the command line or a web browser. Each container runs independently, providing isolated environments to practice various security techniques.

Accessing Kali Linux in command user interface: Kali Linux is an industry-standard platform for penetration testing, offering a wide range of security tools. To interact with the Kali Linux container, you can use the following command in the Windows command prompt:

```
docker exec -it kali /bin/sh
```

This command opens a shell inside the Kali Linux container, allowing you to run tools like nmap, Hydra, and Metasploit. Since Kali Linux is running in a container, you can practice penetration testing without worrying about damaging your host system.

Accessing Kali Linux in graphical user interface through the browser:

We can access the Kali Desktop GUI through any web browser available in Windows at <https://localhost:7002>.

Login credentials for Kali Linux are: Username: kasm_user, Password: password

Accessing DVWA in graphical user interface through browser: DVWA provides a safe environment to practice web application vulnerabilities. You can access the DVWA interface through any web browser available in Windows at <http://localhost:7001>.